Krish Bagga

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Professional Summary

Software engineer with hands-on experience building AWS cloud-native applications, ML pipelines, and DevOps automation including a serverless system supporting 10,000+ concurrent users and a YOLOv5-based document processing pipeline. Skilled in Docker, Kubernetes, CI/CD, Infrastructure as Code, and system design, with proven ability to architect, train, and deploy end-to-end solutions.

Experience

BlackBox IP Corporation | Machine Learning Engineer Intern ***june 15th 26- july 30th 2026***

Python | YOLOv5 | OpenCV | PaddleOCR | PyTorch | Pandas | PDF Processing | CSV/JSON Reporting | CI/CD | Kubernetes

YOLOv5 pipeline:

- Engineered and deployed a YOLOv5-based object detection pipeline to auto-locate inventory name fields in USPTO OATH PDFs, achieving 0.991 precision and 0.929 recall, improved data extraction accuracy by 45% compared to legacy template-matching system.

OCR pipeline upgrade

- Re-architected a legacy OCR/template-matching pipeline with singleton OCR initialization and in-memory template caching, automating batch processing with an 88.7% success rate and CSV/JSON reporting with automatic failed-document routing

Projects

CloudGuard AI – Intelligent AWS Security Platform ***february - april 2026***

Docker | Python | AWS Lambda | FastAPI | Machine Learning | PostgreSQL | REST APIs| jenkins

- Delivered enterprise-grade cloud security features including automated risk scoring, financial impact estimation, and actionable remediation achieving 100% severity classification accuracy at $0 infrastructure cost

- Developed an intelligent AWS security monitoring platform that detects infrastructure misconfigurations across S3, EC2, RDS, and IAM; implemented automated risk scoring, severity classification, and actionable remediation achieving 100% accuracy at $0 infrastructure cost.

Attendees - Automated Attendance & Meal Monitoring using AI Facial Recognition ***October 2025 - November 2025***

AWS Lambda | DynamoDB | Serverless Architecture | API Gateway | CloudWatch | Auto-scaling | jenkins | Docker | GitHub

- Designed and deployed a fully serverless AWS architecture supporting 10,000+ concurrent users using Lambda, API Gateway, DynamoDB with automated scaling; achieved <35ms API latency through optimized NoSQL schema design and application component orchestration.

- Reduced manual attendance workload by 95% through secure auto-scaling architecture with <35ms API latency, automated notifications, and optimized NoSQL schema design.

Research paper

AI Agent-Based Disaster Management & Recovery Systems ***february 2026 - april 2026***

• Conducted research on AI agent-based systems for disaster management and recovery, exploring predictive analytics, automated responseworkflows, and scalable coordination for emergency situations.

Technical skills

Languages: C ++ | Python | SQL | HTML/CSS

Frameworks & Tools: TensorFlow | Scikit-learn | Flask | Selenium | FastAPI

Cloud & DevOps: AWS |Azure | Terraform | Jenkins | Docker | Kubernetes | CI/CD | Linux | Git/GitHub

Technical Skills: Data Structures & Algorithms | Cloud Computing | Machine Learning | Microservices | REST APIs | System Design

Other Skills: Problem Solving | Technical Content Writing | Public Speaking | Team Leadership | Mentorship | Communication Skills

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science Engineering (Cloud Computing Specialization) 2023 – 2027

Bennett University • CGPA: 7.26/10

Certifications

Getting Started with Accelerated Computing in Modern CUDA C++

- WroteparallelGPU-accelerated C++code
- Optimized memory and thread performance
- **Outcome:** Accelerated compute-heavy workloads efficiently

AWS Academy Graduate – Cloud Architecting (60 Hours)

- Designed scalablecloudarchitectures
- Secured and managed AWS services
- **Outcome:** Certified to architect production AWS systems

Extra curriculum

- alogorithmic challenges on LeetCode,codeforces CodeChef
- working in a multi agent env
- constantly upgrading myself through technology trends
- Automating DevOps workflows and scalable deployments